Deep Hucks or Small Ball?

Common Offensive Possession Patterns Among Four 2026 UFA Semifinalists

Team-level spatial analysis of UFA play-by-play data

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Abstract

Professional ultimate possessions are often summarized by their endpoint: a goal, a turnover, or a break. That summary leaves out the route. This paper examines the repeated spatial routes used by the New York Empire, Austin Sol, Minnesota Wind Chill, and Oakland Spiders during the 2026 regular season. We combine throw-level Shown Space data with manually curated field-path groups, then describe each selected pattern using frequency, goal/turnover efficiency, total aECper possession, aECper throw, and throw count. The result is a team-centered view of offensive identity. The same outcome can emerge from a short, aggressive sequence, a longer reset-heavy possession, or a low-huck path that steadily compresses the field. A league-wide throw-count heatmap provides the broader context: five throws is the modal length of an O-linegoal possession, but the four teams examined here occupy noticeably different parts of that distribution.

1 Introduction

The modern UFA analytics conversation has moved well beyond the final line in a box score. Shown Space was introduced as a way to use the league's play-by-play infrastructure to ask how possessions develop, which actions change scoring probability, and which players drive that change (**?**). That shift matters because two possessions can finish in the same end zone while asking very different questions of an offense. One may win through immediate field position and a deep release; another may survive several resets before a cut finally opens the goal line.

The Sloan work behind Shown Space established a related foundation by combining machine learning with completion probability and field value to estimate player decision-making in professional ultimate (**?**). The public articles that followed use that foundation as a storytelling device. In the six-passer comparison, for example, aEC is paired with complementary measures so that distinct jobs remain visible rather than being flattened into one ranking (**?**). The lesson transfers naturally from players to teams: a leaderboard can identify who is productive, while paths can help explain what that productivity looks like.

This paper applies that idea to the four UFA semifinalists represented in the current possession browser. The question is deliberately narrower than "Which team is best?" We ask: *What are the recurring spatial forms of each team's offense, and how do their outcomes and aEC values differ?* The answer should not be reduced to a single style label. The UFA's own team analysis emphasizes that the O-line and D-line can become meaningfully different teams after the disc changes hands, and that a deep ball is a tool rather than a complete offensive identity (**?**). Our analysis therefore stays with O-linepossessions, includes both goals and turnovers, and treats spatial organization as evidence to be inspected rather than a substitute for judgment.

2 Data and Methods

2.1 Sample and possession reconstruction

The analysis uses the cached per-game Shown Space throw files in this project. The regular-season window runs from April 24 through July 19, 2026. The cache also contains later playoff files, so the figure-building script excludes games dated after July 19 before calculating any count, efficiency, or aEC statistic. Each focus team contributes 12 regular-season games. The resulting reconstructed O-linesamples are 257 possessions for New York, 285 for Austin, 266 for Minnesota, and 266 for Oakland.

A possession is formed by grouping throws with the same game, quarter, point, possession number, and home/away side. The final throw determines the outcome: the two retained outcomes are 'goal' and 'turnover'. The line type comes from the source `o_line` field. For every possession, the pipeline retains the ordered throw path, starting and ending field coordinates, throw count, field progress, huck count, reset count, throw-level aEC, and the derived possession totals.

The word "efficiency" has a deliberately local meaning in this paper. For a group of reconstructed O-linepossessions, it is

$$\text{efficiency} = \frac{\text{goals}}{\text{goals} + \text{turnovers}}.$$

It is therefore an outcome ratio for this sample, not a claim that it is identical to the UFA's published OE or expectation-adjusted OEOE measures. Keeping that distinction explicit is essential when comparing a custom browser view with a public leaderboard.

2.2 Human-in-the-loop spatial grouping

The first three rows in each saved arrangement checkpoint (`data/arrangements/2026/empire.json`, `sol.json`, `windchill.json`, and `spiders.json`) are treated as that team's three selected recurring patterns. The grouping was made by reviewing field cards and overlaying paths, then deciding which routes were similar enough to share a row. This is a manual taxonomy with a reproducible checkpoint, not an unsupervised claim that the game contains exactly three natural clusters.

For the paper, a card contributes to a pattern's statistics only when its possession is in the regular-season data and its line type is O-line. The field figures draw every retained path in a selected group, with goals in blue and turnovers in red. That choice keeps the display honest: a group can look spatially coherent while still containing different outcomes.

2.3 Metrics

For a pattern g with n_g possessions and T_g total throws, we report:

- **frequency**: n_g/N_t, where N_t is the team's regular-season O-linepossession count;
- **efficiency**: the goal fraction defined above, with turnovers in the denominator;
- **total aECper possession**: $\sum aEC/n_g$;
- **aECper throw**: $\sum aEC/T_g$; and
- **median throws**: the median number of throws in the pattern.

The throw-level aECvalues are taken from the cached Shown Space rows and summed within each possession. We do not normalize a pattern to equal one or discard negative values. As a result, a successful possession often lands near one total aEC, while a turnover can pull the total below zero; the average is intended to preserve that outcome-sensitive signal. Reporting both total aECper possession and aECper throw also prevents a short possession from being mistaken for a high-volume one.

3 Results

3.1 The league context: five throws is common, not destiny

Figure 1 shows the throw-count distribution for regular-season O-linegoal possessions across all 22 cached teams. Each row is normalized within team, so the darkest cell answers a within-team question: which throw count appears most often among that team's goals? Across the league, the mode is five throws among 3,607 O-linegoal possessions. The focus teams do not share that mode. Austin peaks at four throws; Oakland and Minnesota peak at eight; New York peaks at nine. A mode describes the most frequent length, not an offensive instruction. It is best read alongside the rest of the row, especially when two nearby counts have similar mass.

3.2 Team-level baseline

The four teams separate before the individual patterns are considered (Table 1). New York's reconstructed O-lineconverted 180 of 257 possessions, or 70.0 percent. Oakland followed at 63.5 percent, while Minnesota and Austin finished at 61.3 and 60.0 percent, respectively. New York also led this local sample in average total aECper possession (0.592) and aggregate aECper throw (0.070). Those values should be interpreted as descriptive summaries of the cached rows, not as a replacement for the public Shown Space team metrics.

Table 1: Regular-season reconstructed O-linebaseline.

Team	Games	Pos.	Goals	TOs	Eff.	aEC/pos.	aEC/throw
New York Empire	12	257	180	77	70.0%	0.592	0.070
Austin Sol	12	285	171	114	60.0%	0.431	0.052
Minnesota Wind Chill	12	266	163	103	61.3%	0.486	0.056
Oakland Spiders	12	266	169	97	63.5%	0.492	0.065

3.3 New York Empire: deep options inside a highly efficient base

The Empire's first selected pattern is also its largest of the three selected rows. After the regular-season and O-linefilters, it contains 25 possessions: 23 goals and two turnovers, for 92.0 percent efficiency. Its median length is six throws, and 60.0 percent of the possessions contain a huck. The second and third patterns are smaller, but both converted every retained possession. Pattern 2 is the most compressed value profile of the three, with 0.224 aECper throw across a five-throw median. Pattern 3 reaches the highest average total aECper possession (1.049) while taking a nine-throw median.

That combination suggests an offense with multiple ways to finish a full-field attack. The short pattern is not merely a quick score, and the longer pattern is not automatically less valuable. They differ in how densely value is created, which is precisely why total aECand aECper throw belong in the same conversation. This echoes the broader UFA description of New York as a high- completion offense whose strength comes from the quality of its decisions, not from avoiding every aggressive option (**?**).

3.4 Austin Sol: a fast branch and two longer answers

Austin's three selected rows reveal a different balance. Pattern 1 is the shortest, with a four-throw median and a 100 percent huck rate, but its 11 possessions converted at 72.7 percent. Pattern 2 is more frequent (19 possessions) and longer (10 throws at the median), while Pattern 3 begins from a more advanced field position and has the highest efficiency of the three at 83.3 percent. The longer two

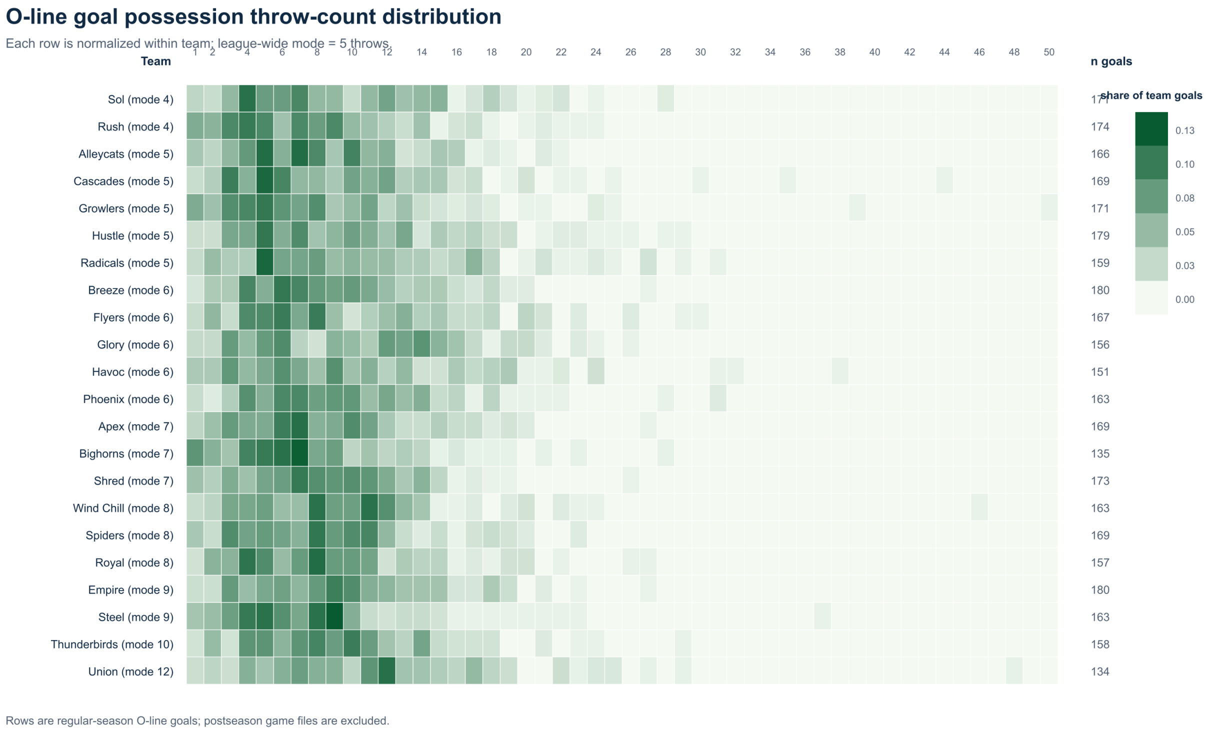

Figure 1: Regular-season throw-count distribution for O-linegoal possessions. Color is the share of each team's O-linegoals, not the share of all league possessions. The right-hand count is the number of goal possessions used for that row.

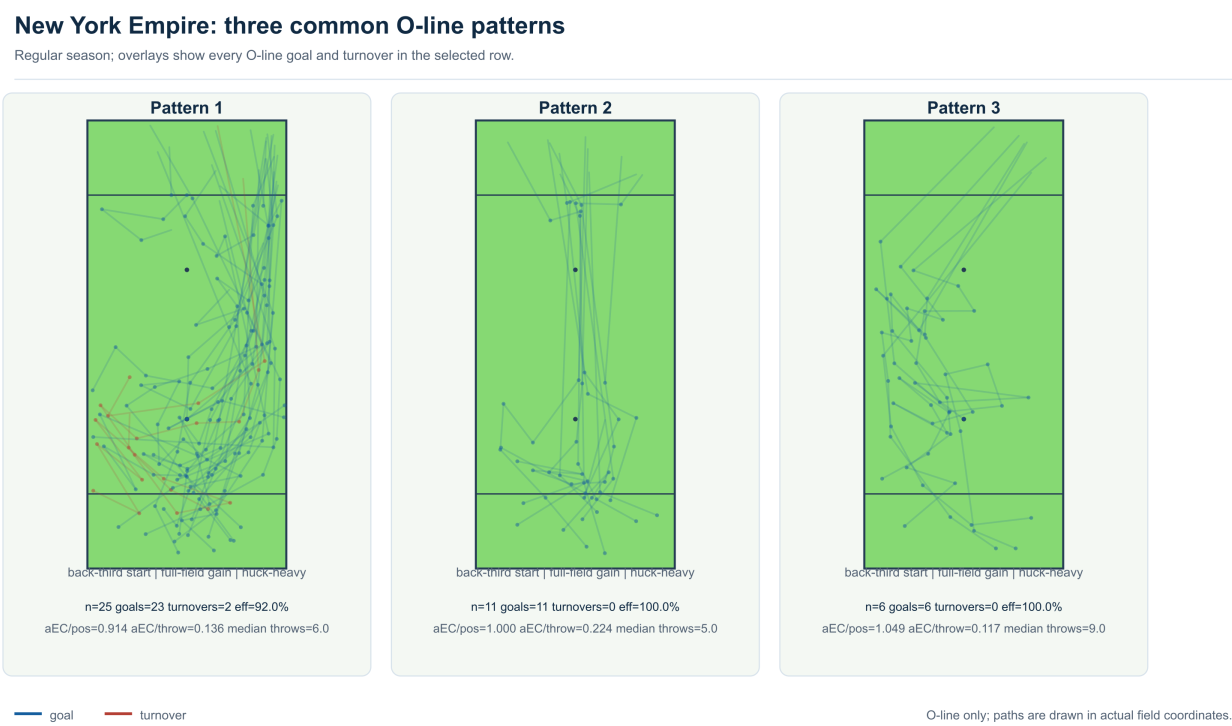

Figure 2: New York Empire's three selected recurring patterns. Each field overlays the regular-season O-linepaths retained in that row; blue is a goal and red is a turnover. The metrics below each field include both outcomes.

patterns account for 37 selected regular-season possessions and together show that a Sol attack can keep working after the first high-value option disappears.

The aEC numbers reinforce that contrast. Pattern 1 creates value most densely (0.132 per throw), whereas Pattern 3 produces the largest average total among the three (0.876 per possession). A single ranking would make one of those facts disappear. The public UFA writing makes a similar move when it separates team identity from a single efficiency number and emphasizes how a unit gets to its finishing chances (?).

3.5 Minnesota Wind Chill: a repeatable middle with small high-value branches

Minnesota's first pattern is the meaningful base of its selected set: 17 possessions, an eight-throw median, 76.5 percent efficiency, and 0.671 total aECper possession. Patterns 2 and 3 are much smaller, with five and two possessions, respectively, and both finished without a turnover in this sample. Their perfect rates should be read as evidence about the observed cards, not as stable estimates of the team's true conversion probability. Pattern 3's 1.071 total aECper possession is attractive, but it is supported by only two regular-season possessions.

The spatial story is therefore one of a common middle path surrounded by specialized branches. Minnesota's modal goal length is eight throws, and the selected patterns mostly sit near eight to ten throws. That aligns with the broader observation that modern UFA offenses can trade frequent deep attempts for longer possessions and more deliberate progression (?).

3.6 Oakland Spiders: low-huck continuity and width

Oakland's first two selected patterns contain no hucks at all. Pattern 1 has eight goals and one turnover across nine possessions; Pattern 2 has 12 goals and one turnover across 13. Their median lengths are ten and eleven throws, respectively. Pattern 3, titled "lateral up left sideline" in the saved arrangement, is shorter at an eight-throw median and converted all ten retained possessions. Its 0.121 aECper throw is the highest density among Oakland's three selected patterns.

This is a useful example of why "small ball" should not be confused with "short." The first two Oakland patterns are low-huck, but they are not brief. They move through a sequence of catches and throws while using the width of the field. That reading is consistent with the public team profile, which describes the Spiders as an offense that uses the full width of the field and pairs that with a carefully selected deep game (?). The field overlays make the distinction visible in a way that a huck rate alone cannot.

3.7 The comparison in one table

Table 2 places the selected rows beside one another. The numbers show why frequency, efficiency, and aEC should be allowed to disagree. Empire's first row is its largest selected group, while the smaller second row has the highest aECper throw in the entire four-team set. Austin's best conversion rate belongs to its third row, not its most frequent one. Oakland's first two groups combine strong conversion with long, low-huck sequences. And Minnesota's two perfect rows are too small to carry the same evidentiary weight as its 17-possession base pattern.

4 Discussion

4.1 What the patterns add to the leaderboard

The main contribution of these displays is interpretive rather than merely decorative. A possession path supplies a bridge between a metric and a tactical description. The six-passer article makes that bridge

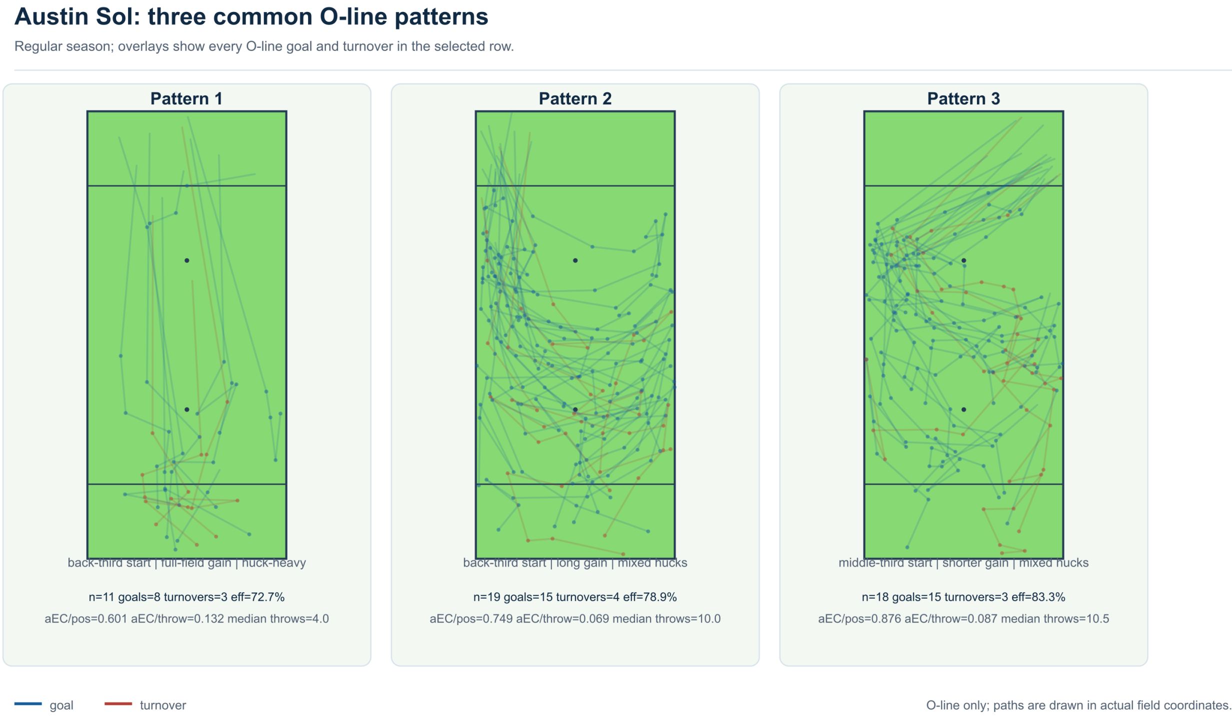

Figure 3: Austin Sol's three selected recurring patterns. The selected rows include both goal and turnover paths; the sample size beneath each field keeps the apparent 100 percent or near-100 percent rates in context.

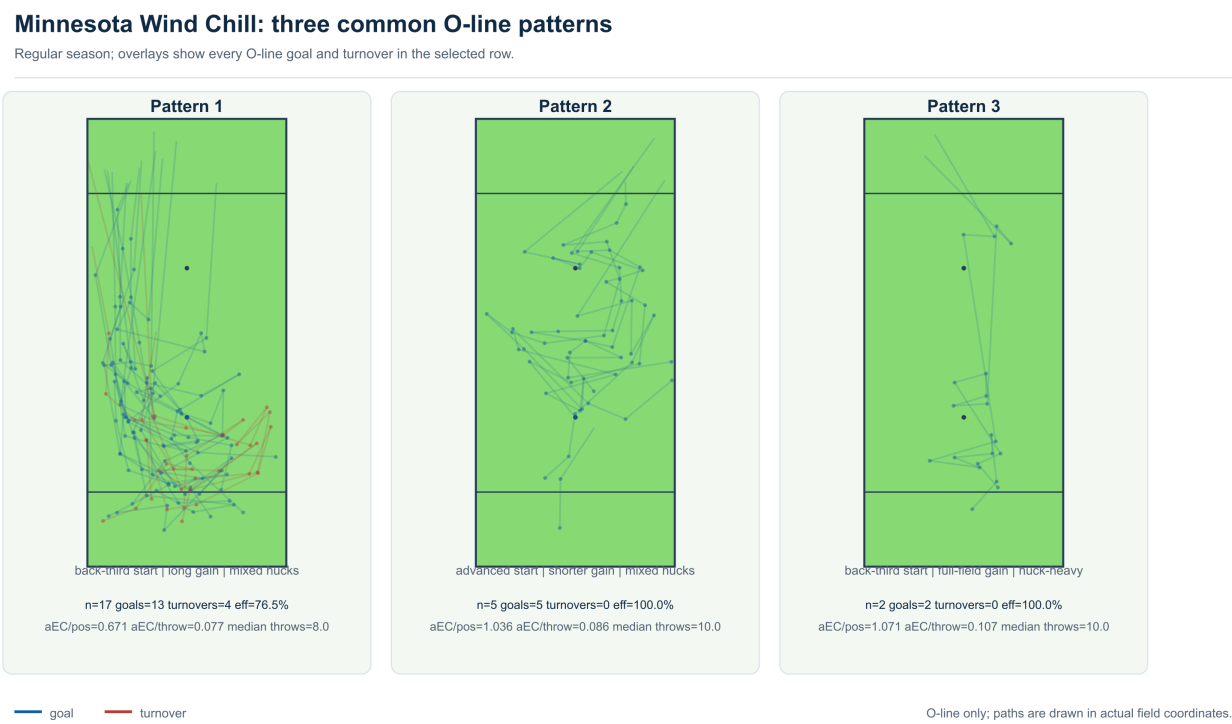

Figure 4: Minnesota Wind Chill's three selected recurring patterns. Pattern 3 is deliberately shown with its small n rather than being allowed to read as a general team tendency.

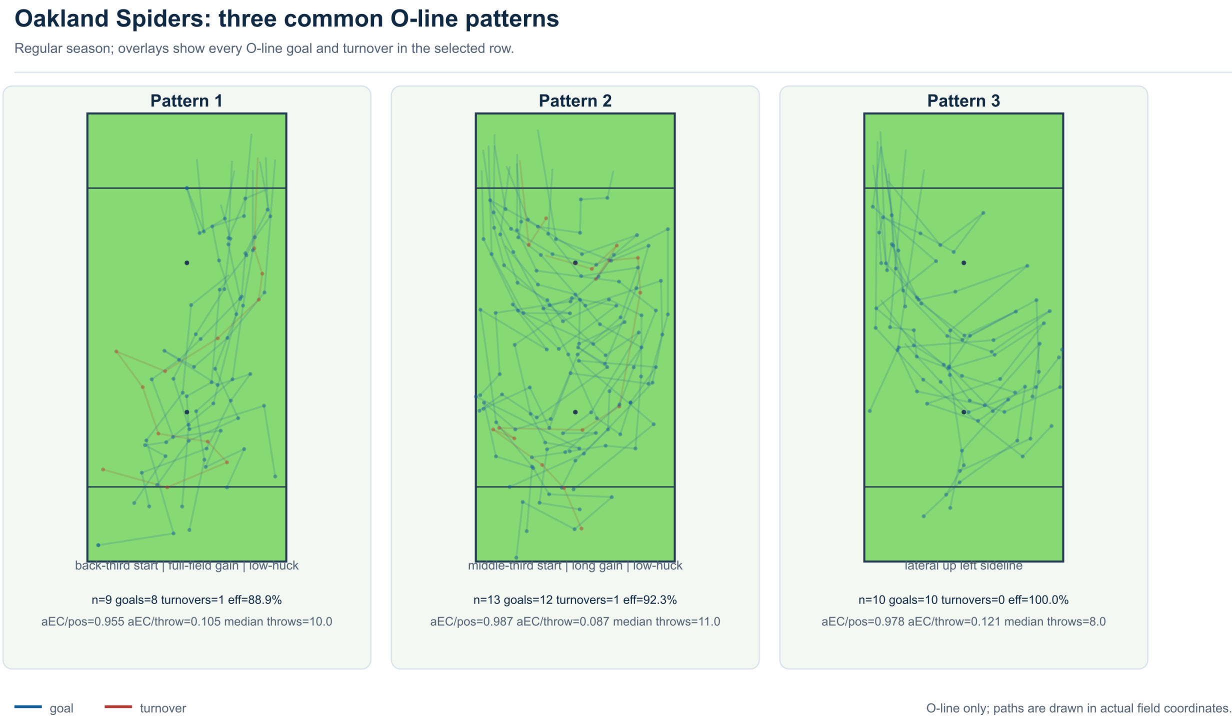

Figure 5: Oakland Spiders' three selected recurring patterns. The first two rows are low-huck, longer sequences; the third is the saved lateral-up-left- sideline group.

Table 2: Metrics for the three selected patterns per focus team. Efficiency includes turnovers; aEC/pos. is the mean total aEC, and aEC/throw is the aggregate total aEC divided by aggregate throws.

Team	Pat.	n	Share	G	TO	Eff.	aEC/pos.	aEC/throw	Med. throws
New York Empire	1	25	9.7%	23	2	92.0%	0.914	0.136	6.0
New York Empire	2	11	4.3%	11	0	100.0%	1.000	0.224	5.0
New York Empire	3	6	2.3%	6	0	100.0%	1.049	0.117	9.0
Austin Sol	1	11	3.9%	8	3	72.7%	0.601	0.132	4.0
Austin Sol	2	19	6.7%	15	4	78.9%	0.749	0.069	10.0
Austin Sol	3	18	6.3%	15	3	83.3%	0.876	0.087	10.5
Minnesota Wind Chill	1	17	6.4%	13	4	76.5%	0.671	0.077	8.0
Minnesota Wind Chill	2	5	1.9%	5	0	100.0%	1.036	0.086	10.0
Minnesota Wind Chill	3	2	0.8%	2	0	100.0%	1.071	0.107	10.0
Oakland Spiders	1	9	3.4%	8	1	88.9%	0.955	0.105	10.0
Oakland Spiders	2	13	4.9%	12	1	92.3%	0.987	0.087	11.0
Oakland Spiders	3	10	3.8%	10	0	100.0%	0.978	0.121	8.0

at the player level by showing that high aEC can come from very different combinations of shooting, volume, field vision, or support (**?**). Here, the same logic operates one level up. Empire's selected rows suggest several efficient full-field solutions. Oakland's rows suggest continuity through space without needing frequent hucks. Austin and Minnesota show how a longer possession can be a recurring branch rather than a failure of the first option.

The distinction between total aECand aECper throw is especially useful. Total aECasks how much value the entire possession accumulated. aECper throw asks how concentrated that value was across the sequence. A short possession can look excellent on the latter measure even when another longer possession produces more total value. Neither is a universal quality score; each answers a different question about the route.

4.2 Turnovers change the pattern story

If only scoring possessions were retained, several of the selected overlays would look cleaner and several efficiency numbers would become meaningless. Including turnovers changes the interpretation in two ways. First, it makes "how often this pattern scores" an observable outcome ratio rather than a description of successful examples only. Second, it lets spatially similar routes reveal where the same idea breaks down. The red paths in the figures are not clutter; they are the negative evidence needed to distinguish a reliable pattern from a memorable one.

That is also why the manuscript reports n beside every percentage. A 100 percent group with two possessions is a promising observation. It is not a league-ready estimate. The public Shown Space writing repeatedly combines volume, efficiency, risk, and role when telling player stories (**??**); team patterns deserve the same discipline.

4.3 From recurring paths to tactical language

The current labels should remain modest. "Lateral up left sideline" is a geometric description grounded in the cards. A claim such as "Oakland always attacks the left sideline" would require more evidence: force direction, opponent quality, personnel, game state, and the paths that did not score. Likewise, a five-throw mode does not prove that a team is trying to reach five throws. It simply identifies the most common observed length in the selected outcome set.

The productive next step is a second manual pass. Once a team has a small set of spatial families, an analyst can name them using both their geometry and their function: for example, a direct deep branch, a lateral continuation sequence, or a long reset-and-advance pattern. The figures make those judgments auditable. Another analyst can see the same overlays, disagree with a boundary, and revise the group without changing the underlying possession reconstruction.

5 Limitations

Several limitations should shape how these results are used.

First, the pattern taxonomy is analyst-curated. The three rows are useful descriptive units, but their boundaries reflect visual judgment and the current arrangement checkpoints. A future version should measure inter-rater agreement or compare the manual rows with a geometry-first clustering baseline.

Second, the paper studies four teams and three selected rows per team. Large unsorted groups remain outside the narrative by design. The figures are not a complete description of any offense, and they should not be used to infer that an unshown pattern is rare without checking the full browser.

Third, the sample is regular season only. It does not answer whether a team changed its offense in the playoffs. Nor does the current summary adjust for opponent pressure, field conditions, score state, personnel, or starting field position. These factors can influence both the path and the outcome.

Finally, the paper uses the cached throw-level aECvalues supplied by Shown Space; it does not refit or audit the underlying machine-learning model. The metric is treated as a contextual contribution signal, not as a causal estimate that a pattern alone produced the final score.

6 Conclusion

The four teams in this study do not reduce to a single offensive template. Empire's selected groups combine high conversion with multiple efficient full-field routes. Austin carries a short, aggressive branch alongside longer sequences. Minnesota's common middle is surrounded by smaller high-value branches whose uncertainty is visible once turnovers and sample size are shown. Oakland's selected patterns make the strongest case for low-huck continuity: long sequences, high conversion, and a spatial identity that depends on width more than on constant deep shots.

The practical result is a way to write about team offense that stays close to the field. The heatmap says how long possessions tend to last. The overlays say where the disc travels. Efficiency and aECsay what those routes produced. Together they turn a possession browser into an argument about how teams build scores, while leaving enough of the evidence visible for the reader to make a different judgment.

Reproducibility

All calculations and figures in this manuscript are regenerated with `scripts/build_paper_figures.py`. The script reads the four saved arrangements, filters the cached game files through July 19, 2026, writes the CSV/LaTeX tables under `paper/generated/`, and produces standalone SVG figures. The arrangement checkpoints remain separate from the generated figures so later manual edits can be compared with this version. The SVG figures are converted to the checked-in PNGs by `scripts/render_paper_figures.cjs` when the figures are regenerated.